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COVID-19 pandemic deep learning implementations of prediction of disease with data analysis and real-time face-mask detection with camera

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***Abstract*—In biomedical sciences, data mining skills are used to research and provide predictions to aid in the identification and classification of diseases. Controlling the spread of Corona Virus Disease requires screening a high number of reported cases for effective isolation and treatment (COVID-19). Infective laboratory testing (Pathogenic) is the benchmark in science, but it is time-consuming because of the high rate of false-negative?findings. To treat the illness, there is an urgent need for rapid and dependable diagnosis techniques.We wanted to create a deep learning system that could retrieve COVID-19 pictorial features from Computed tomography applying COVID-19 radiographic enhancements. In earlier study investigations, machine learning methods were employed in the prediction and categorization of COVID-19. This research, on the other hand, concentrates on the different effects of certain image processing techniques rather than on optimising these processes through the use of improved approaches. The CT image dataset benefits from the extraction of classified correctness. The DeTraC model, a previously published convolutional neural network architecture based on class decomposition, is used in this study to increase the performance of pre-trained models in detecting COVID-19 instances from chest X-ray pictures. This may be accomplished by including a class breakdown layer into the pre-trained models.**
Keywords: Deep Learning, feature extraction, VGG19, CNN, exploratory data analysis, visualization, ResNet, etc.

I. Introduction

The emerging coronavirus illness 2019 (COVID-19) pandemic produced by SARS-CoV-2 poses a serious and urgent threat to global health. The outbreak began in early December 2019 in the People's Republic of China's Hubei province and has since expanded worldwide. This pandemic is still posing numerous problems to medical systems throughout the world, including huge increases in demand for hospital beds and acute shortages of medical equipment, while many healthcare personnel have gotten ill.

As a consequence, the emphasis on making rapid medical decisions and successfully manage pharmaceutical resources is crucial. The most reliable COVID-19 diagnostic method, reverse transcriptase-P.C.R. refers for polymerase chain reaction (RT-PCR), has always been scarce in developing countries. This adds to higher infection rates and causes essential preventative actions to be delayed. Comprehensive screening enables the fast and effective recognition of COVID-19, decreasing the burden on the medical system. Prediction models that integrate numerous characteristics to predict the risk of infection have been created in the hopes of aiding medical professionals throughout the globe in triaging patients, particularly in the setting of limited healthcare resources. Computer tomography (CT) images are used in these simulations. In this work, we introduce a machine-learning algorithm that, using Lung X-ray images of the chest, predicts a positive COVID-19 infection.

II. Motivation

The COVID-19 Coronavirus epidemic has produced a crisis all across the world. COVID-19's cumulative incidence is quickly growing day by day. We can utilised Machine Learning to detect the disease extremely well, and Exploratory Data Analysis can be used for Data Visualization. We can utilise Machine Learning to manage enormous amounts of data and diagnose diseases intelligently. The major goal was to reduce the disease and assist in the detection because of the illness and the face mask with the greatest precision.

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III. Related Work

Digital technology have played critical roles in significant health sector issues such as illness prevention since the previous decade. In order to tackle COVID-2019, the current international health crisis is also seeking technological help. We illustrated the new digital technologies such as big data analytics, artificial intelligence (AI), blockchain technology, internet of things (IOT), and deep learning in formulating policies for epidemic tracking, identification, and prevention, as well as evaluating the consequence of the epidemic on the medical industry [1].

Benvenuto et al, [2] They used an autoregressive integrated moving average (ARIMA) technique to estimate the transmission of COVID-2019 in their academic publication. Based on a study of the prevalence and incidence of COVID-2019, the author predicted the different metrics in this article over the next two days. This study also includes a correlogram and an ARIMA forecast graph for pandemic prevalence and incidence. Deb and colleagues [3] presented a time series approach for analysing COVID-19 outbreak incidence patterns and the predicted reproduction number.

They used statistical analysis to investigate the patterns of the epidemic in order to emphasise the current epidemiological stage of an area so that alternative approaches to handle the COVID-19 pandemic in different nations may be recognised [4]. According to the current circumstances, In order to build and regulate suitable protective measures, it is vital to understand the illness's earliest transmission characteristics. Kucharski et al. used a scientific model of critical SARS-CoV2 transmission as well as several datasets to analyse the COVID-19 outbreak inside and outside Wuhan city of china. They used this to investigate the probable transmission of illness epidemics outside of Wuhan. [5].

Numerous research on the COVID-19 epidemiological pandemic have recently been performed, utilising exploratory analysis of the data (EDA) on a range of available to the public datasets. The research primarily focus on the incidence of confirmed, fatal, and recovered cases in Wuhan and throughout the world in order to understand the suspected hazards and plan future containment efforts [13]. Lauer et al. emphasised the issue of the significance of the COVID-19 incubation period in their study.

IV. Proposed Framework

This study employs a Convolutional Neural Network (CNN) to perform image classification tests on X-ray pictures of individuals as input datasets, in order to determine whether or not a specific person is COVID-19 positive. And we apply VGG19, our deep learning technique for image classification, to determine whether or not A person is covered in a mask. Furthermore, the statistical data given might be utilised to conduct additional study.

A. Convolutional Neural Network

By training the CNN model using the image dataset and then applying it to video classification, which requires examining a batch of frames from the input video to determine its class, the suggested technique reduces computational costs while improving accuracy. As seen in the figure 1, a video is used as an input to our model.

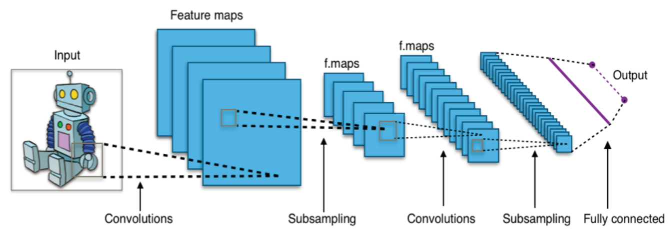

Fig. 1. CNN model architecture

The video is made up of frames, with each frame having an input dimension of 224*224 pixels. If the picture does not fit the specified dimensions, the model will only be trained to take the dimensions listed above into consideration. Video categorization entails more than simply classifying images. Because the succeeding frames in a video are connected to their semantic contents, the suggested model would be efficient because it uses the same type of data as the traditional model.However, as stated in the objective, the CNN's performance is to be evaluated, and the same model is to be trained on the Keras library, with further predictions made using the Rolling Prediction Approach, which is detailed in the next chapter.

B. VGG 19

The VGG research group produced a series of convolution network models from VGG11 to VGG19. The VGG depth group's main objective is to investigate how the depth of convolutional neural networks affects the accuracy of large - scale image classification and recognition models. VGG11 has eight convolutional layers and three fully connected layers, whereas VGG19 has sixteen convolutional layers and 3 fully connected layers.

In the last three completely linked layers, the various variants of VGGs are the same. The overall structure consists of five sets of convolutional layers, then comes a MaxPool. Moreover, as the level increases, from VGG11 to VGG19, even more spiralled convolutional layers are introduced to the five sets of convolutional layers.

C. ResNet

A neural network with residuals (ResNet) is a kind of artificial neural network (ANN) that is based on prefrontal cortical pyramidal neurons. Skip connections are accomplished through the use of Residual neural networks., or shortcuts, to bypass some layers. In most cases, ResNet models include double- sometimes triple-layer skips with nonlinearities and batch normalisation in between. An additional weight matrix can be used to learn the skip weights; these models have been found as HighwayNets. DenseNets are models with a large number of parallel skips.

A plain network is a non-residual network, Whereas in the setting of residual neural networks. A pyramidal cell reconstruction. The soma is highlighted in red also dendrites are labelled in red, whereas axon arbour is labelled in blue. Basal, Soma dendrite, Apical dendrite, Axon and Collateral axon. One is to avoid the problem of vanishing gradients, and the other is to avoid the Degradation (accuracy saturation) problem, which happens when adding extra layers to a suitably deep model leads in increased training mistake, These are two key reasons for including skip connections.

The weights fluctuate while training to mute the upstream layer and highlight the previously skipped layer. In the most basic instance, only the weights for the link to the adjacent layer are revised, with no explicit weights for the upstream layer. When stepping over a single nonlinear layer or when the intervening layers are all linear, this approach works well. In the alternative, an explicit weight matrix for the missing connection should be learned (a HighwayNet should be used).

V. Experiments and results

A. Dataset

The system will be fed three different datasets. The first is a numerical dataset taken through web scrapping from government website [16], followed by an chest X-ray image dataset which is taken from Kaggle and Git-Hub Repository [17] [18], and finally a mask images dataset from Kaggle [19]. These datasets are all accessible on Kaggle. A numerical dataset is used for real-time data visualisation through exploratory data analysis and web scraping. The Lung X-ray image data collection is used to predict COVID-19 illness. The mask dataset is used for real-time mask detection. Our system's output will take three possible forms:

- In Real time Data Visualization
- In Real time COVID-19 Disease Prediction
- In Real time Determine whether or not A person is covered in a mask.

The proposed method is to minimise computing costs while improving accuracy by training the CNN using the image dataset and then utilising it for video classification, which involves checking a batch of frames from the input video to identify its class. However, as stated in the objective, the CNN's performance will be evaluated, and the same model will be trained on the Keras library and additional predictions will be made using the Rolling Prediction Approach.

B. Training

We use web scraping in our system to get data from the Indian Government website. First, we use a third-party HTTP library for python-requests to submit an web(HTTP) request to the website in order to obtain the content. After gaining access, we parse the data with the html5lib parser to create a kind of tree construction of the HTML data. We next explore and lookup the parse tree with a third-party Python package called Beautiful Soup before extracting data from that tree and appending it to our dataset.

We utilised CNN for COVID-19 prediction (Convolutional Neural Network). We evaluated the accuracy of ResNet18 and CNN and found that CNN had the greatest accuracy of 92 Percent, therefore we selected CNN as our model for COVID-19 detection. The Training Set contains 11045 pictures with the class labels "COVID and NON-COVID." We add more Convolution layers, as well as a Maxpooling layer, and a dropout layer. Flatten a 2D image layer to create a 1D image. For the amount of nodes and activation, we built a thick layer. In our system, 0.5 drop out rates are utilised, half of the input nodes are dropped at each update, and 30 epoch cycles are utilized in the training of model. In the event of hazy inputs, our updated Convolutional Neural Network model readily recognises and detects the outcomes.

The trials are carried out on a computer equipped with a graphics processing unit (GPU),Core i5 processor of Intel corporation or higher 7th generation 2.4 GHz CPU, and RAM must be at least 8 GB. A minimum of 240 GB hard drive space and 8 GB RAM with a clock speed of 2 GHz or higher are required. The essential prerequisites for Web application development with MySQL include 4GB of RAM.

C. System Architecture

The architecture of the system is the most appropriate way to displaying the overall structure. For the initial training of the CNN model, which is done just once, input can be provided in the form of pictures. The model then predicts whether the person is covid positive or normal based on the lung x-ray picture. VGG19 is used by the face mask detection module to determine whether or not a person is covered with a mask. Real-time EDA is accomplished through the use of visualisation tools, and data for EDA is obtained by web scraping.

D. Result

TABLE I
Accuracy table

Accuracy Comparison		
Model Name	Previous paper	This paper
CNN	0.92	0.94
VGG-19	-	0.98

The suggested model's final findings are shown in table 1, including phase-wise outputs and figures. The model has been validated with several X-ray pictures of the lungs. The requested model has been coded and implemented effectively. The model that resulted has been tested on various traffic conditions and has worked well. The project that has been executed provides an accurate and timely answer to the problem of traffic congestion. The project also meets all of its objectives.

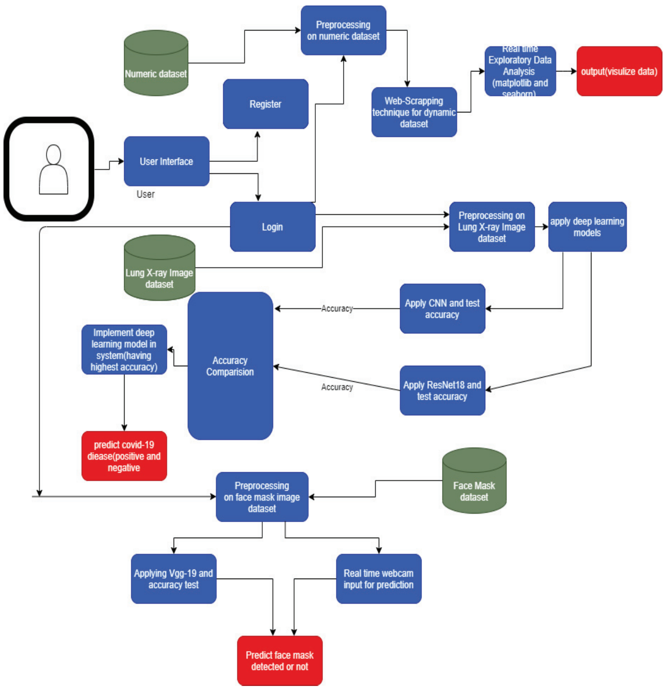

Fig. 2. System Architecture

The examination of the Indian COVID-19 dataset to estimate the number of confirmed cases, actual cases, fatalities, and recovered cases is demonstrated in Figure 3. Figure 4 depicts the detection of the mask with VGG 19 algorithm. Figure 5 depicts COVID-19 prediction using a CNN model.

VI. Conclusion

COVID-19 diagnosis is generally linked with COVID-19 symptoms, which can be disclosed by genetic and imaging examinations. The Imagine test can enable rapid diagnosis of COVID-19 and so help to limit disease transmission. The imaging methods CXR and CT are crucial in the detection of COVID-19 infection.

Because of the provision of labelled large-scale picture datasets, significant progress has been achieved in deep CNNs for medical image classification. CNNs allow for the direct learning of local picture characteristics that are highly representational and hierarchical from data. However, inconsistencies in annotated data continue to be the most difficult problem in dealing with genuine COVID-19 cases from CXR pictures. The benefit is that verifying individuals affected is critical to properly managing and containing the infection. It would be difficult to ascertain the exact frequency of instances without proper testing. As a result, it is critical to understand what these accessible tests can and cannot accomplish in order to apply them effectively.

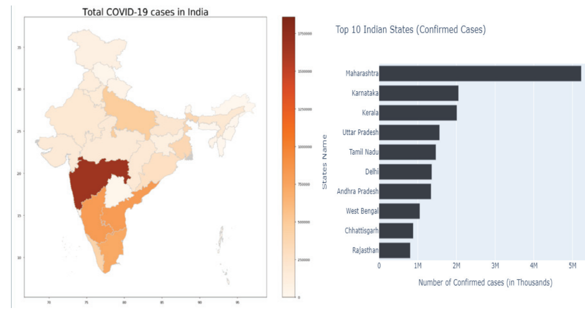

Fig. 3. Statistical Analysis

Fig. 4. Mask Detection

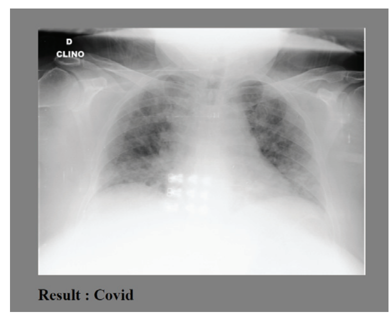

Fig. 5. COVID-19 Prediction

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